

-1 Notation

$$K = \mathrm{GL}(2, \mathbb{O}_F), I = \left\{ \kappa = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K : c \equiv 0 \pmod{\varpi \mathbb{O}_F} \right\}.$$

$$f_{2k} = \frac{1}{\mathrm{Vol}} 1_{K_k}, K_k = \begin{pmatrix} 1 + \varpi^k \mathbb{O}_F & \varpi^k \mathbb{O}_F \\ \varpi^k \mathbb{O}_F & 1 + \varpi^k \mathbb{O}_F \end{pmatrix}.$$

$$f_{4k+1} = \frac{1}{\mathrm{Vol}} 1_{U_k}, U_k = \begin{pmatrix} 1 + \varpi^k \mathbb{O}_F & \varpi^k \mathbb{O}_F \\ \varpi^{k+1} \mathbb{O}_F & 1 + \varpi^k \mathbb{O}_F \end{pmatrix}.$$

$$f_{4k-1} = \frac{1}{\mathrm{Vol}} 1_{V_k}, V_k = \begin{pmatrix} 1 + \varpi^k \mathbb{O}_F & \varpi^{k-1} \mathbb{O}_F \\ \varpi^k \mathbb{O}_F & 1 + \varpi^k \mathbb{O}_F \end{pmatrix}.$$

$$\mathbb{O}_E \subset K, \mathbb{O}_E = \mathbb{O}_F + \theta \mathbb{O}_F.$$

$$L_r = \mathbb{O}_F + \theta \varpi^r \mathbb{O}_F.$$

We will calculate the following integral.

$$I(s, x) = \sum_{r \geq 0} \left(\int_K \tilde{f} \left(\kappa \begin{pmatrix} \varpi^{-r} & \\ & 1 \end{pmatrix} x \begin{pmatrix} \varpi^r & \\ & 1 \end{pmatrix} \kappa^{-1} \right) d\kappa \right) \mathrm{wt}(s; r).$$

Here we choose $\tilde{f}$ to be our test function calculated from Jacquet-Langlands correspondence.

0 Preparation

By definition we have the following equalities.

$$\frac{\mathrm{Vol}(K)}{\mathrm{Vol}(K_k)} = q^{4k-3}(q-1)^2(q+1).$$

$$\frac{\mathrm{Vol}(I)}{\mathrm{Vol}(U_k)} = q^{4k-2}(q-1)^2.$$

$$\frac{\mathrm{Vol}(I)}{\mathrm{Vol}(V_k)} = q^{4k-4}(q-1)^2.$$

By the calculation before, we have the following orbital integrals.

$$\int_K f_{2k} \left(\kappa \begin{pmatrix} \varpi^{-r} & \\ & 1 \end{pmatrix} x \begin{pmatrix} \varpi^r & \\ & 1 \end{pmatrix} \kappa^{-1} \right) d\kappa = \frac{\mathrm{Vol}(K)}{\mathrm{Vol}(K_k)} \cdot 1_{1+\varpi^k L_r}(x).$$

$$\int_K f_{4k-1} \left(\kappa \begin{pmatrix} \varpi^{-r} & \\ & 1 \end{pmatrix} x \begin{pmatrix} \varpi^r & \\ & 1 \end{pmatrix} \kappa^{-1} \right) d\kappa = \frac{\mathrm{Vol}(I)}{\mathrm{Vol}(V_k)} \cdot \begin{cases} q \cdot 1_{1+\varpi^k L_r}(x) + 1_{1+\varpi^k L_{r-1}}(x), & (\text{for all } r > 0) \\ (q+1) \cdot 1_{1+\varpi^k \mathbb{O}_E}(x), & (r=0, E/F \text{ unramified}) \\ q \cdot 1_{1+\varpi^k \mathbb{O}_E}(x) + 1_{1+\varpi_E^{2k-1} \mathbb{O}_E}(x). & (r=0, E/F \text{ ramified}) \end{cases}$$

$$\int_K f_{4k+1} \left(\kappa \begin{pmatrix} \varpi^{-r} & \\ & 1 \end{pmatrix} x \begin{pmatrix} \varpi^r & \\ & 1 \end{pmatrix} \kappa^{-1} \right) d\kappa = \frac{\mathrm{Vol}(I)}{\mathrm{Vol}(U_k)} \cdot \begin{cases} q \cdot 1_{1+\varpi^k L_{r+1}}(x) + 1_{1+\varpi^k L_r}(x), & (\text{for all } r > 0) \\ (q+1) \cdot 1_{1+\varpi^k L_1}(x), & (r=0, E/F \text{ unramified}) \\ q \cdot 1_{1+\varpi^k L_1}(x) + 1_{1+\varpi^k \mathbb{O}_E}(x). & (r=0, E/F \text{ ramified}) \end{cases}$$

By Proposition 4.5, we have the following equalities.

$$\mathrm{wt}(s; r) = \begin{cases} (1+q^{-1})^{-1}, & (r > 0, E/F \text{ unramified}) \\ 1, & (r > 0, E/F \text{ ramified}) \\ \frac{(Z-q^{-1}Z^{-1})(q^{\frac{1}{2}}Z)^r - (Z^{-1}-q^{-1}Z)(q^{\frac{1}{2}}Z^{-1})^r}{Z-Z^{-1}}, & (r=0, E/F \text{ unramified}) \\ \frac{(Z-q^{-\frac{1}{2}})(q^{\frac{1}{2}}Z)^r - (Z^{-1}-q^{-\frac{1}{2}})(q^{\frac{1}{2}}Z^{-1})^r}{Z-Z^{-1}}. & (r=0, E/F \text{ ramified}) \end{cases}.$$

Z stands for q^s .

1 **E/F unramified**, $\tilde{f} = -\frac{2}{q-1}f_{2k} + \frac{q+1}{q-1}f_{4k-1}$

For $r > 0$,

$$\int_K \tilde{f} \left(\kappa \begin{pmatrix} \varpi^{-r} & \\ & 1 \end{pmatrix} x \begin{pmatrix} \varpi^r & \\ & 1 \end{pmatrix} \kappa^{-1} \right) d\kappa = q^{4k-4}(q-1)(q+1) \cdot \left(-q \cdot 1_{1+\varpi^k L_r}(x) + 1_{1+\varpi^k L_{r-1}}(x) \right).$$

For $r = 0$,

$$\int_K \tilde{f} \left(\kappa \begin{pmatrix} \varpi^{-r} & \\ & 1 \end{pmatrix} x \begin{pmatrix} \varpi^r & \\ & 1 \end{pmatrix} \kappa^{-1} \right) d\kappa = q^{4k-4}(q-1)(q+1) \cdot \left(-(q-1) \cdot 1_{1+\varpi^k O_E}(x) \right).$$

Let

$$\tilde{I}(s, x) = I(s, x) \cdot (q^{4k-4}(q-1)(q+1))^{-1}.$$

$$\begin{aligned} \tilde{I}(s, x) &= -\frac{q-1}{1+q^{-1}} \cdot 1_{1+\varpi^k O_E}(x) + \sum_{r>0} \left(-q \cdot 1_{1+\varpi^k L_r} + 1_{1+\varpi^k L_{r-1}}(x) \right) \cdot \frac{(Z - q^{-1}Z^{-1})(q^{\frac{1}{2}}Z)^r - (Z^{-1} - q^{-1}Z)(q^{\frac{1}{2}}Z^{-1})^r}{Z - Z^{-1}} \\ &= I_0 1_{1+\varpi^k O_E}(x) + \sum_{r>0} I_r 1_{1+\varpi^k L_r}(x). \end{aligned}$$

First we calculate I_0 .

$$\begin{aligned} I_0 &= -\frac{q(q-1)}{q+1} + \frac{(q^s - q^{-1-s})q^{\frac{1}{2}+s} - (q^{-s} - q^{s-1})q^{\frac{1}{2}-s}}{q^s - q^{-s}} \\ &= -\frac{q(q-1)}{q+1} + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}). \end{aligned}$$

Then we calculate I_r .

$$\begin{aligned} I_r &= \frac{(Z - q^{-1}Z^{-1})(q^{\frac{1}{2}}Z)^r(q^{\frac{1}{2}}Z - q) - (Z^{-1} - q^{-1}Z)(q^{\frac{1}{2}}Z^{-1})^r(q^{\frac{1}{2}}Z^{-1} - q)}{Z - Z^{-1}} \\ &= q \frac{(q^{1+2s} - 1)q^{-1-s}q^{(\frac{1}{2}+s)r}(q^{s-\frac{1}{2}} - 1) - (q^{1-2s} - 1)q^{-1+s}q^{(\frac{1}{2}-s)r}(q^{-s-\frac{1}{2}} - 1)}{q^s - q^{-s}} \\ &= -q^{-\frac{1}{2}} \frac{(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}+s} + 1)q^{(\frac{1}{2}+s)r}(q^{\frac{1}{2}-s} - 1) - (q^{\frac{1}{2}-s} - 1)(q^{\frac{1}{2}-s} + 1)q^{(\frac{1}{2}-s)r}(q^{\frac{1}{2}+s} - 1)}{q^s - q^{-s}} \\ &= -q^{-\frac{1}{2}}(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} - 1) \frac{q^{(\frac{1}{2}+s)r}(q^{\frac{1}{2}+s} + 1) - q^{(\frac{1}{2}-s)r}(q^{\frac{1}{2}-s} + 1)}{q^s - q^{-s}}. \end{aligned}$$

Now if $(x-1) \in \varpi^k L_{r_x} \setminus \varpi^k L_{r_x+1}$, then we can write $\tilde{I}(s, x)$ as a function of r_x and s .

$$\begin{aligned}
\tilde{I}(s, x) &= \sum_{r=0}^{r_x} I_r \\
&= -\frac{q(q-1)}{q+1} + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}) - q^{-\frac{1}{2}}(q^{\frac{1}{2}-s} - 1)(q^{\frac{1}{2}+s} - 1) \sum_{r=1}^{r_x} \frac{q^{(\frac{1}{2}+s)r}(q^{\frac{1}{2}+s} + 1) - q^{(\frac{1}{2}-s)r}(q^{\frac{1}{2}-s} + 1)}{q^s - q^{-s}}. \\
&= -\frac{q(q-1)}{q+1} + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}) - q^{-\frac{1}{2}} \frac{(q^{\frac{1}{2}-s} - 1)(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}+s} + 1)(q^{(\frac{1}{2}+s)(r_x+1)} - q^{\frac{1}{2}+s})}{(q^s - q^{-s})(q^{\frac{1}{2}+s} - 1)} \\
&\quad + q^{-\frac{1}{2}} \frac{(q^{\frac{1}{2}-s} - 1)(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} + 1)(q^{(\frac{1}{2}-s)(r_x+1)} - q^{\frac{1}{2}-s})}{(q^s - q^{-s})(q^{\frac{1}{2}-s} - 1)} \\
&= -\frac{q(q-1)}{q+1} + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}) - \frac{(q^{\frac{1}{2}-s} - 1)(q^{\frac{1}{2}+s} + 1)q^s}{q^s - q^{-s}}(q^{(\frac{1}{2}+s)r_x} - 1) \\
&\quad + \frac{(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} + 1)q^{-s}}{q^s - q^{-s}}(q^{(\frac{1}{2}-s)r_x} - 1) \\
&= -\frac{q(q-1)}{q+1} + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}) + q^{\frac{1}{2}r_x} \cdot \left(-q^{sr_x} \frac{(q^{\frac{1}{2}-s} - 1)(q^{\frac{1}{2}+s} + 1)q^s}{q^s - q^{-s}} + q^{-sr_x} \frac{(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} + 1)q^{-s}}{q^s - q^{-s}} \right) \\
&\quad + \left(\frac{(q^{\frac{1}{2}-s} - 1)(q^{\frac{1}{2}+s} + 1)q^s}{q^s - q^{-s}} - \frac{(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} + 1)q^{-s}}{q^s - q^{-s}} \right) \\
&= \frac{q-1}{q+1} + q^{\frac{1}{2}r_x} \cdot \left(-q^{sr_x} \frac{(q^{\frac{1}{2}-s} - 1)(q^{\frac{1}{2}+s} + 1)q^s}{q^s - q^{-s}} + q^{-sr_x} \frac{(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} + 1)q^{-s}}{q^s - q^{-s}} \right).
\end{aligned}$$

There are two remarkable special values for $\tilde{I}(s, x)$.

First we consider $s = 1/2$. Direct calculation shows that

$$\tilde{I}\left(\frac{1}{2}, x\right) = \frac{q-1}{q+1} + q^{\frac{1}{2}r_x} \left(-q^{\frac{1}{2}r_x} \frac{(q^{\frac{1}{2}-\frac{1}{2}} - 1)(q^{\frac{1}{2}+\frac{1}{2}} + 1)q^{\frac{1}{2}}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}} + q^{-\frac{1}{2}r_x} \frac{(q^{\frac{1}{2}+\frac{1}{2}} - 1)(q^{\frac{1}{2}-\frac{1}{2}} + 1)q^{-\frac{1}{2}}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}} \right) = \frac{q-1}{q+1} + 2 = \frac{3q+1}{q+1}.$$

Next we consider $s = 0$.

$$\begin{aligned}
\tilde{I}(0, x) &= \frac{q-1}{q+1} + q^{\frac{1}{2}r_x} \lim_{s \rightarrow 0} \left(\frac{-q^{sr_x}(q^{\frac{1}{2}} - q^s)(q^{\frac{1}{2}+s} + 1) + q^{-sr_x}(q^{\frac{1}{2}} - q^{-s})(q^{\frac{1}{2}-s} + 1)}{q^s - q^{-s}} \right). \\
&= \frac{q-1}{q+1} + q^{\frac{1}{2}r_x} \lim_{s \rightarrow 0} \frac{q^{\frac{1}{2}}(q^{s(r_x+2)} - q^{-s(r_x+2)}) - (q-1)(q^{s(r_x+1)} - q^{-s(r_x+1)}) - q^{\frac{1}{2}}(q^{s(r_x+2)} - q^{-s(r_x+2)})}{q^s - q^{-s}} \\
&= \frac{q-1}{q+1} + q^{\frac{1}{2}r_x} (q^{\frac{1}{2}}(r_x + 2) - (q-1)(r_x + 1) - q^{\frac{1}{2}r_x})
\end{aligned}$$

2 E/F ramified, $\tilde{f} = -\frac{2}{q-1}f_{2k} + \frac{q+1}{q-1}f_{4k-1}$

For $r > 0$,

$$\int_K \tilde{f} \left(\kappa \begin{pmatrix} \varpi^{-r} & \\ & 1 \end{pmatrix} x \begin{pmatrix} \varpi^r & \\ & 1 \end{pmatrix} \kappa^{-1} \right) d\kappa = q^{4k-4}(q-1)(q+1) \cdot (-q \cdot 1_{1+\varpi^k L_r}(x) + 1_{1+\varpi^k L_{r-1}}(x)).$$

For $r = 0$,

$$\int_K \tilde{f} \left(\kappa \begin{pmatrix} \varpi^{-r} & \\ & 1 \end{pmatrix} x \begin{pmatrix} \varpi^r & \\ & 1 \end{pmatrix} \kappa^{-1} \right) d\kappa = q^{4k-4}(q-1)(q+1) \cdot (-q \cdot 1_{1+\varpi^k O_E}(x) + 1_{1+\varpi^{2k-1} O_E}(x)).$$

Let

$$\tilde{I}(s, x) = I(s, x) \cdot (q^{4k-4}(q-1)(q+1))^{-1}.$$

$$\begin{aligned}\tilde{I}(s, x) &= -q \cdot 1_{1+\mathfrak{w}^k 0_E}(x) + 1_{1+\mathfrak{w}_E^{2k-1} 0_E}(x) + \sum_{r>0} \left(-q \cdot 1_{1+\mathfrak{w}^k L_r} + 1_{1+\mathfrak{w}^k L_{r-1}}(x) \right) \cdot \frac{(Z - q^{-\frac{1}{2}})(q^{\frac{1}{2}}Z)^r - (Z^{-1} - q^{-\frac{1}{2}})(q^{\frac{1}{2}}Z^{-1})^r}{Z - Z^{-1}} \\ &= I_0 1_{1+\mathfrak{w}^k 0_E}(x) + 1_{1+\mathfrak{w}_E^{2k-1} 0_E}(x) + \sum_{r>0} I_r 1_{1+\mathfrak{w}^k L_r}(x).\end{aligned}$$

First we calculate I_0 .

$$\begin{aligned}I_0 &= -q + \frac{(q^s - q^{-\frac{1}{2}})q^{\frac{1}{2}+s} - (q^{-s} - q^{-\frac{1}{2}})q^{\frac{1}{2}-s}}{q^s - q^{-s}} \\ &= -(q+1) + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}).\end{aligned}$$

Then we calculate I_r .

$$\begin{aligned}I_r &= \frac{(Z - q^{-\frac{1}{2}})(q^{\frac{1}{2}}Z)^r(q^{\frac{1}{2}}Z - q) - (Z^{-1} - q^{-\frac{1}{2}})(q^{\frac{1}{2}}Z^{-1})^r(q^{\frac{1}{2}}Z^{-1} - q)}{Z - Z^{-1}} \\ &= q^{\frac{1}{2}} \frac{(q^{s+\frac{1}{2}} - 1)q^{(\frac{1}{2}+s)r}(q^{s-\frac{1}{2}} - 1) - (q^{-s+\frac{1}{2}} - 1)q^{(\frac{1}{2}-s)r}(q^{-s-\frac{1}{2}} - 1)}{q^s - q^{-s}} \\ &= -(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} - 1) \frac{q^{(\frac{1}{2}+s)r}q^s - q^{(\frac{1}{2}-s)r}q^{-s}}{q^s - q^{-s}}.\end{aligned}$$

Now if $(x-1) \in \mathfrak{w}^k L_{r_x} \setminus \mathfrak{w}^k L_{r_x+1}$, then we can write $\tilde{I}(s, x)$ as a function of r_x and s .

$$\begin{aligned}\tilde{I}(s, x) &= 1 + \sum_{r=0}^{r_x} I_r \\ &= -q + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}) - (q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} - 1) \sum_{r=1}^{r_x} \frac{q^{(\frac{1}{2}+s)r}q^s - q^{(\frac{1}{2}-s)r}q^{-s}}{q^s - q^{-s}} \\ &= -q + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}) - \frac{(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} - 1)q^s(q^{(\frac{1}{2}+s)(r_x+1)} - q^{(\frac{1}{2}+s)})}{(q^s - q^{-s})(q^{\frac{1}{2}+s} - 1)} \\ &\quad + \frac{(q^{\frac{1}{2}+s} - 1)(q^{\frac{1}{2}-s} - 1)q^{-s}(q^{(\frac{1}{2}-s)(r_x+1)} - q^{(\frac{1}{2}-s)})}{(q^s - q^{-s})(q^{\frac{1}{2}-s} - 1)} \\ &= -q + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}) - \frac{(q^{\frac{1}{2}-s} - 1)q^s(q^{(\frac{1}{2}+s)(r_x+1)} - q^{(\frac{1}{2}+s)})}{q^s - q^{-s}} + \frac{(q^{\frac{1}{2}+s} - 1)q^{-s}(q^{(\frac{1}{2}-s)(r_x+1)} - q^{(\frac{1}{2}-s)})}{q^s - q^{-s}} \\ &= -q + (q^{\frac{1}{2}+s} + q^{\frac{1}{2}-s}) + \frac{(q^{\frac{1}{2}-s} - 1)q^{\frac{1}{2}+2s} - (q^{\frac{1}{2}+s} - 1)q^{\frac{1}{2}-2s}}{q^s - q^{-s}} \\ &\quad + q^{\frac{1}{2}(r_x+1)} \cdot \left(-q^{s(r_x+1)} \frac{(q^{\frac{1}{2}-s} - 1)q^s}{q^s - q^{-s}} + q^{-s(r_x+1)} \frac{(q^{\frac{1}{2}+s} - 1)q^{-s}}{q^s - q^{-s}} \right) \\ &= q^{\frac{1}{2}(r_x+1)} \cdot \left(-q^{s(r_x+1)} \frac{(q^{\frac{1}{2}-s} - 1)q^s}{q^s - q^{-s}} + q^{-s(r_x+1)} \frac{(q^{\frac{1}{2}+s} - 1)q^{-s}}{q^s - q^{-s}} \right).\end{aligned}$$

There are two remarkable special values for $\tilde{I}(s, x)$.

First we consider $s = 1/2$. Direct calculation shows that

$$\tilde{I}(\frac{1}{2}, x) = q^{\frac{1}{2}(r_x+1)} \left(-q^{\frac{1}{2}(r_x+1)} \frac{(q^{\frac{1}{2}-\frac{1}{2}} - 1)q^{\frac{1}{2}}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}} + q^{-\frac{1}{2}(r_x+1)} \frac{(q^{\frac{1}{2}+\frac{1}{2}} - 1)q^{-\frac{1}{2}}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}} \right) = 1.$$

Next we consider $s = 0$. By L'Hôpital rule, we have

$$\begin{aligned} \tilde{I}(0, x) &= q^{\frac{1}{2}(r_x+1)} \lim_{s \rightarrow 0} \frac{-q^{s(r_x+1)}(q^{\frac{1}{2}} - q^s) + q^{-s(r_x+1)}(q^{\frac{1}{2}} - q^{-s})}{q^s - q^{-s}}. \\ &= q^{\frac{1}{2}(r_x+1)} \lim_{s \rightarrow 0} \frac{-(r_x+1) \log q \cdot q^{s(r_x+1)}(q^{\frac{1}{2}} - q^s + q^{\frac{1}{2}} - q^{-s}) + q^s \log q (q^{s(r_x+1)} + q^{-s(r_x+1)})}{(q^s + q^{-s}) \cdot \log q} \\ &= q^{\frac{1}{2}(r_x+1)} \cdot (1 - (r_x+1)(q^{\frac{1}{2}} - 1)) = (r_x+1)q^{\frac{1}{2}(r_x+1)} \left(q^{\frac{1}{2}} - \left(1 + \frac{1}{r_x+1} \right) \right). \end{aligned}$$

3 Riemann hypothesis for $\tilde{I}(s, x)$, E/F ramified

Obviously $\tilde{I}(s, x) = \tilde{I}(-s, x)$. Since we have such symmetry, we expect some analogue of Riemann hypothesis.

Lemma (unit circle theorem, version I). Let n be any positive integer, and $t > 1 + 1/n$ a real number. Consider the polynomial

$$f(Z) = Z^{2n+2} - t(Z^{2n+1} - Z) - 1,$$

then the roots of $f(Z)$ must be one of the following two cases.

1. On the unit circle, for $2n$ of them.
2. On the positive real line but not 1, for two of them. The product of the two roots should be 1.

Proof. We first prove $f(Z) = 0$ have $2n$ roots on the unit circle. Let $Z = e^{i\alpha}$, where $\alpha \in \mathbb{R}$. It is obvious that

$$f(Z) = 0, Z \text{ on the unit circle} \Leftrightarrow \Xi(\alpha) = \sin((n+1)\alpha) - t \cdot \sin(n\alpha) = 0.$$

Consider $\alpha_m = (m+1/2)\pi/n$, here m takes value in integers. And since When m is an even number we know

$$\Xi(\alpha_m) = \sin((n+1)\alpha_m) - t \cdot \sin(n\alpha_m) \leq 1 - t < 0.$$

When m is an odd number we know

$$\Xi(\alpha_m) = \sin((n+1)\alpha_m) - t \cdot \sin(n\alpha_m) \geq -1 + t > 0.$$

Using intermediate value theorem, exists $\tilde{\alpha}_m \in (\alpha_m, \alpha_{m+1})$, such that $\Xi(\tilde{\alpha}_m) = 0$. Define $Z_m = e^{i\tilde{\alpha}_m}$. By our definition we know $Z_0, Z_1, \dots, Z_{2n-1}$ are distinct roots of $f(Z)$.

Then we prove $f(Z) = 0$ have 2 roots on the positive real line but not 1. Let $Z = e^\beta$, where $\beta \in \mathbb{R}$. It is obvious that

$$f(Z) = 0, Z > 0 \text{ but not } 1 \Leftrightarrow \upsilon(\beta) = \frac{\sinh((n+1)\beta)}{\sinh(n\beta)} = t.$$

Take first derivative we have

$$\frac{d\upsilon}{d\beta}(\beta) = \frac{1}{2} (\sinh((2n+1)\beta) - (2n+1) \sinh(\beta)) \operatorname{csch}^2(\beta).$$

By classical inequality $\sinh(ax) > a \cdot \sinh(x)$ we know when $\beta > 0$ we have $\frac{d\upsilon}{d\beta} > 0$. And since when $\beta \rightarrow +\infty$, $\upsilon(\beta) \rightarrow +\infty$. Also $\lim_{\beta \rightarrow 0} \upsilon(\beta) = 1 + 1/n$. Thus, exists a unique $\beta_c \in (0, +\infty)$ such that $\upsilon(\beta_c) = t$. By the evenness, $\upsilon(-\beta_c) = t$. Then $e^{-\beta_c}, e^{\beta_c}$ are also roots of $f(Z)$.

Since $f(Z)$ is a degree $2n+2$ polynomial, so we have exhausted all possible roots of $f(Z)$. $\square$

Remark. If $0 < t < 1 + 1/n$, then all $2n+2$ roots should lie on the unit circle. If $t = 1 + 1/n$, then not only all $2n+2$ roots all lie on the unit circle, but also we have 1 to be a triple root of $f(Z)$.

Theorem (RH for $\tilde{I}(s, x)$, E/F ramified). $\tilde{I}(s, x)$ is defined as in section 2. Fixing x , all zeros of $\tilde{I}(s, x)$ lie on the vertical line $\text{Re}(s) = 0$ or on $\pm s_c + 2\pi i\mathbb{Z}$, except for $(q, r_x) = (2, 1)$. Here s_c is the unique real positive root of $\tilde{I}(s, x)$.

Proof. In the proof we write $Z = q^s$. From now on we assume $s \neq 0$.

$$\tilde{I}(s, x) = 0 \Leftrightarrow 0 = -Z^{r_x+1}(q^{\frac{1}{2}} - Z) + Z^{-(r_x+1)}(q^{\frac{1}{2}} - Z^{-1}) \Leftrightarrow f(Z) = 0.$$

Here we take $f(Z) = Z^{2r_x+4} - q^{\frac{1}{2}}(Z^{2r_x+3} - Z) - 1$. $q^{\frac{1}{2}} > 1 + \frac{1}{r_x+1}$ holds unless $(q, r_x) = (2, 1)$.

However, $s = 0$ is not a zero of $\tilde{I}(s, x)$, since $q^s - q^{-s}$ has a first order zero at $s = 0$. $\square$

Theorem. $s_c < 1/2$.

Proof. Let $f(Z) = Z^{2r_x+4} - q^{\frac{1}{2}}(Z^{2r_x+3} - Z) - 1$. Easily seen $f(q^{1/2}) = q - 1 > 0$. $\square$

4 Riemann Hypothesis for $\tilde{I}(s, x)$, E/F unramified

Similarly we have $\tilde{I}(s, x) = \tilde{I}(-s, x)$.

Lemma (unit circle theorem, version II). Let n be any positive integer, and $t > C$ a real number. Consider the polynomial

$$f(Z) = \frac{t^2 - 1}{t^2 + 1}(Z^{n+3} - Z^{n+1}) + t^n \cdot (tZ^{2n+4} - (t^2 - 1)Z^{2n+3} - tZ^{2n+2} + tZ^2 + (t^2 - 1)Z - t),$$

then the roots of $f(Z)$ must be one of the following two cases.

1. On the unit circle, for $2n+2$ of them.
2. On the positive real line but not 1, for two of them. The product of the two roots should be 1.

Here C is an absolute constant.

Sketch of the proof. We first prove $f(Z) = 0$ have $2n+2$ roots on the unit circle. Let $Z = e^{i\alpha}$, where $\alpha \in \mathbb{R}$. It is obvious that

$$f(Z) = 0, Z \text{ on the unit circle}$$

$$\Leftrightarrow \Xi(\alpha) = \frac{t^2 - 1}{t^2 + 1} \sin(\alpha) + t^n \cdot (t \sin((n+2)\alpha) - (t^2 - 1) \sin((n+1)\alpha) - t \sin(n\alpha)) = 0.$$

Here we can use the same argument in the version I, as long as

$$t^2 - 1 > 2t + \frac{t^2 - 1}{t^2 + 1} \cdot t^{-n}.$$

Then we prove $f(Z) = 0$ have 2 roots on the positive real line but not 1. The key point is to determine when $f(Z) = 0$ has a triple root, which as we see earlier, can be derived from the sign of $f(Z)$ near $Z = 1$. The tipping point t_c satisfy a transcendental equation:

$$2 - (t_c^2 - 1)(n+1) + \frac{t_c^2 - 1}{t_c^2 + 1} t_c^{-n} = 0.$$

If $n \gg 1$, we can get an asymptotic formula:

$$t_c = 1 + \frac{1}{n} + O\left(\frac{1}{n^2}\right).$$

The formula is similar to the ramified case. $\square$

Theorem (RH for $\tilde{I}(s, x)$, E/F unramified). $\tilde{I}(s, x)$ is defined as in section 2. Fixing x , all zeros of $\tilde{I}(s, x)$ lie on the vertical line $\operatorname{Re}(s) = 0$ or on $\pm s_c + 2\pi i\mathbb{Z}$, with a finite number of exceptions. Here s_c is the unique real positive root of $\tilde{I}(s, x)$.

The proof is as in the ramified case.

Theorem. $s_c < 1/2$.

Proof. Let $f(Z) = \frac{t^2-1}{t^2+1}(Z^{n+3} - Z^{n+1}) + t^n \cdot (tZ^{2n+4} - (t^2 - 1)Z^{2n+3} - tZ^{2n+2} + tZ^2 + (t^2 - 1)Z - t)$. Here $t = q^{1/2}$. Then we know

$$f(q^{1/2}) = f(t) = t^{n+1}(t^2 - 1) \left(2 + \frac{t^2 - 1}{t^2 + 1} \right) = q^{\frac{n+1}{2}}(q - 1) \frac{3q + 1}{q + 1} > 0.$$

□